

Homework 2

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1 (2A-2). Prove or give a counterexample: If v_1, v_2, v_3, v_4 span V , then the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans V .

Proof. We will prove the statement.

By the definition of spanning, we have that for every $v \in V$, there exist scalars a_1, a_2, a_3, a_4 such that

$$v = a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4.$$

Then, we can rewrite the above equation as

$$v = a_1(v_1 - v_2) + (a_1 + a_2)(v_2 - v_3) + (a_1 + a_2 + a_3)(v_3 - v_4) + (a_1 + a_2 + a_3 + a_4)v_4.$$

This means that for every $v \in V$, there exist scalars b_1, b_2, b_3, b_4 such that

$$v = b_1(v_1 - v_2) + b_2(v_2 - v_3) + b_3(v_3 - v_4) + b_4v_4.$$

□

Problem 2 (2A-6). Show that the list $(2, 3, 1), (1, -1, 2), (7, 3, c)$ is linearly dependent in $\mathbb{F}^3$ if and only if $c = 8$.

Proof. On the one hand, if $c = 8$, then we have

$$2(2, 3, 1) + 3(1, -1, 2) - (7, 3, 8) = (0, 0, 0).$$

On the other hand, if the list $(2, 3, 1), (1, -1, 2), (7, 3, c)$ is linearly dependent in $\mathbb{F}^3$, then there exist scalars a_1, a_2, a_3 with $a_3 \neq 0$ (otherwise the first two vectors would be linearly dependent) such that

$$a_1(2, 3, 1) + a_2(1, -1, 2) + a_3(7, 3, c) = (0, 0, 0).$$

This means that

$$\begin{cases} 2a_1 + a_2 + 7a_3 = 0, \\ 3a_1 - a_2 + 3a_3 = 0, \\ a_1 + 2a_2 + ca_3 = 0. \end{cases}$$

Then

$$0 = -7(2a_1 + a_2 + 7a_3) + 3(3a_1 - a_2 + 3a_3) + 5(a_1 + 2a_2 + ca_3) = (5c - 40)a_3.$$

Since $a_3 \neq 0$, we have $c = 8$. □

Problem 3 (2A-7). (a) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{R}$, then the list

$$1 + i, 1 - i$$

is linearly independent.

(b) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{C}$, then the list

$$1 + i, 1 - i$$

is linearly dependent.

Solution. (a) Suppose that there exist scalars $a_1, a_2 \in \mathbb{R}$ such that

$$a_1(1 + i) + a_2(1 - i) = 0.$$

By comparing the real and imaginary parts of the above equation, we have

$$a_1 + a_2 = 0, \quad a_1 - a_2 = 0. \quad \Rightarrow \quad a_1 = a_2 = 0.$$

(b) Note that

$$(1 + i) - i(1 - i) = 0.$$

Problem 4 (2B-4).

(a) Let U be the subspace of $\mathbb{C}^5$ defined by

$$U = \{(z_1, z_2, z_3, z_4, z_5) \in \mathbb{C}^5 : 6z_1 = z_2 \text{ and } z_3 + 2z_4 + 3z_5 = 0\}.$$

Find a basis of U .

(b) Extend the basis in (a) to a basis of $\mathbb{C}^5$.

(c) Find a subspace W of $\mathbb{C}^5$ such that $\mathbb{C}^5 = U \oplus W$.

Solution. (a) We parametrize the subspace U using free variables. From $6z_1 = z_2$, we have $z_2 = 6z_1$. From $z_3 + 2z_4 + 3z_5 = 0$, we have $z_3 = -2z_4 - 3z_5$.

Let $z_1 = s$, $z_4 = t$, $z_5 = u$ be free variables. Then

$$(z_1, z_2, z_3, z_4, z_5) = (s, 6s, -2t - 3u, t, u) = s(1, 6, 0, 0, 0) + t(0, 0, -2, 1, 0) + u(0, 0, -3, 0, 1).$$

Therefore, a basis of U is

$$\{e_1 := (1, 6, 0, 0, 0), e_2 := (0, 0, -2, 1, 0), e_3 := (0, 0, -3, 0, 1)\}.$$

(b) To extend this to a basis of $\mathbb{C}^5$, we add the vectors $e_4 := (0, 1, 0, 0, 0)$ and $e_5 := (0, 0, 1, 0, 0)$. On the one hand, it is easy to verify that the five vectors are linearly independent. On the other hand, for every vector $(z_1, z_2, z_3, z_4, z_5) \in \mathbb{C}^5$, we have

$$(z_1, z_2, \dots, z_5) = z_1 e_1 + z_4 e_2 + z_5 e_3 + (z_1 - 5z_2) e_4 + (z_3 + 2z_4 + 3z_5) e_5.$$

(c) Let $W = \text{span}\{e_4, e_5\}$. Since $U = \text{span}\{e_1, e_2, e_3\}$ and notice that $\{e_1, e_2, \dots, e_5\}$ is a basis of $\mathbb{C}^5$, hence $U \cap W = \{0\}$ and $U + W = \text{span}\{e_1, e_2, e_3, e_4, e_5\} = \mathbb{C}^5$. Therefore, $\mathbb{C}^5 = U \oplus W$.

Problem 5 (2B-5). Suppose V is finite-dimensional and U, W are subspaces of V such that $V = U + W$. Prove that there exists a basis of V consisting of vectors in $U \cup W$.

Proof. Obviously, $U \cap W$ is a subspace of V . Pick a basis $\mathcal{B}_0$ of $U \cap W$; if $U \cap W = \{0\}$, let $\mathcal{B}_0 = \emptyset$. Then we can extend $\mathcal{B}_0$ to a basis $\mathcal{B}_1$ of U and a basis $\mathcal{B}_2$ of W .

We claim that $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$ is a basis of V . On the one hand, for every vector $v \in V$, there exist vectors $u \in U$ and $w \in W$ such that $v = u + w$. Notice that $u \in \text{span}\mathcal{B}_1$ and $w \in \text{span}\mathcal{B}_2$, hence $v \in \text{span}\mathcal{B}$, which implies that $\mathcal{B}$ spans V .

On the other hand, if the vectors in $\mathcal{B}$ are linearly dependent, then there exists a linear combination of vectors in $\mathcal{B}$ whose coefficients are not all zero. Note that $\mathcal{B} = \mathcal{B}_1 \cup (\mathcal{B}_2 - \mathcal{B}_1)$, and $\mathcal{B}_1 \cap (\mathcal{B}_2 - \mathcal{B}_1) = \emptyset$, then the linear combination can be rewritten as a linear combination of vectors in $\mathcal{B}_1$ plus a linear combination of vectors in $\mathcal{B}_2 - \mathcal{B}_1$. In other words, there exist a vector $v_1 \in \text{span}\mathcal{B}_1$ and a vector $v_2 \in \text{span}(\mathcal{B}_2 - \mathcal{B}_1)$ such that $v_1 + v_2 = 0$ and $v_1 \neq 0, v_2 \neq 0$. Then, $v_2 = -v_1 \in \text{span}\mathcal{B}_1$, which indicates $v_2 \in (\text{span}\mathcal{B}_1) \cap (\text{span}(\mathcal{B}_2 - \mathcal{B}_1)) = \{0\}$, which contradicts the fact that $v_2 \neq 0$. Therefore, the vectors in $\mathcal{B}$ are linearly independent.

Hence, $\mathcal{B}$ is a basis of V consisting of vectors in $U \cup W$. $\square$

Problem 6 (2B-11). Suppose V is a real vector space. Show that if $v_1, \dots, v_n$ is a basis of V (as a real vector space), then $v_1, \dots, v_n$ is also a basis of the complexification $V_{\mathbb{C}}$ (as a complex vector space).

Proof. First, we show that the list $v_1, \dots, v_n$ spans $V_{\mathbb{C}}$. For every vector $v \in V_{\mathbb{C}}$, there exist vectors $u, w \in V$ such that $v = u + iw$. Since $v_1, \dots, v_n$ is a basis of V , there exist scalars $a_1, \dots, a_n, b_1, \dots, b_n \in \mathbb{R}$ such that

$$u = a_1 v_1 + \dots + a_n v_n, \quad w = b_1 v_1 + \dots + b_n v_n.$$

Then

$$v = u + iw = (a_1 v_1 + \dots + a_n v_n) + i(b_1 v_1 + \dots + b_n v_n) = (a_1 + ib_1)v_1 + \dots + (a_n + ib_n)v_n.$$

This shows that $v \in \text{span}\{v_1, \dots, v_n\}$, so the list $v_1, \dots, v_n$ spans $V_{\mathbb{C}}$.

Next, we show that the list $v_1, \dots, v_n$ is linearly independent in $V_{\mathbb{C}}$. Suppose that there exist scalars $c_1, \dots, c_n \in \mathbb{C}$ such that

$$c_1 v_1 + \dots + c_n v_n = 0.$$

Write $c_j = a_j + ib_j$ for some $a_j, b_j \in \mathbb{R}$. Then

$$(a_1 + ib_1)v_1 + \dots + (a_n + ib_n)v_n = (a_1 v_1 + \dots + a_n v_n) + i(b_1 v_1 + \dots + b_n v_n) = 0.$$

Since $v_1, \dots, v_n$ is a basis of V , we have $a_1 = \dots = a_n = 0$ and $b_1 = \dots = b_n = 0$. Therefore, $c_1 = \dots = c_n = 0$, which shows that the list $v_1, \dots, v_n$ is linearly independent in $V_{\mathbb{C}}$.

Combining the above two parts, it's proved that $v_1, \dots, v_n$ is a basis of $V_{\mathbb{C}}$. $\square$

Problem 7 (2C-6).

- (a) Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) : p(2) = p(5) = p(6)\}$. Find a basis of U .
- (b) Extend the basis in (a) to a basis of $\mathcal{P}_4(\mathbb{F})$.
- (c) Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Solution. (a) Set

$$\begin{cases} e_1(x) := 1, \\ e_2(x) := (x-2)(x-5)(x-6), \\ e_3(x) := x(x-2)(x-5)(x-6). \end{cases}$$

We will show that $\{e_1, e_2, e_3\}$ is a basis of U .

On the one hand, for every $p \in U$, we have $p(2) = p(5) = p(6)$. Let $c = p(2)$. Then the polynomial $p - ce_1$ has roots 2, 5, 6, which means that there exists a linear function $ax + b$ such that

$$p - ce_1 = (ax + b)(x-2)(x-5)(x-6) \Rightarrow p = ae_3 + be_2 + ce_1.$$

On the other hand, if $ae_3 + be_2 + ce_1 = 0$ for some scalars a, b, c , then by comparing the coefficients of x^4 , we have $a = 0$. Then by comparing the coefficients of x^3 , we have $b = 0$. Finally, by comparing the coefficients of x^0 , we have $c = 0$. Therefore, the vectors in $\{e_1, e_2, e_3\}$ are linearly independent.

Hence, $\{e_1, e_2, e_3\}$ is a basis of U .

(b) Set

$$\begin{cases} e_4(x) := (x-2)(x-5), \\ e_5(x) := (x-2)(x-6). \end{cases}$$

We will show that $\{e_1, e_2, e_3, e_4, e_5\}$ is a basis of $\mathcal{P}_4(\mathbb{F})$. On the one hand, for every polynomial $p \in \mathcal{P}_4(\mathbb{F})$, it is easy to verify that (by taking $x = 2, 3, 4, 5, 6$ and use the Fundamental Algebra Theorem)

$$\begin{aligned} p &= p(2)e_1 + \left(\frac{5}{4}p(2) + \frac{2}{3}p(3) - \frac{3}{4}p(4) - p(5) - \frac{1}{6}p(6)\right)e_2 \\ &\quad + \left(-\frac{3}{4}p(2) - \frac{1}{6}p(3) + \frac{1}{4}p(4) + \frac{1}{2}p(5) + \frac{1}{6}p(6)\right)e_3 + (p(6) - p(2))e_4 + (p(5) - p(2))e_5. \end{aligned}$$

On the other hand, we will show that the vectors in $\{e_1, e_2, e_3, e_4, e_5\}$ are linearly independent. Suppose that there exist scalars a_1, a_2, a_3, a_4, a_5 such that

$$a_1e_1 + a_2e_2 + a_3e_3 + a_4e_4 + a_5e_5 = 0.$$

By comparing the coefficients of x^4 , we have $a_3 = 0$. Then by comparing the coefficients of x^3 , we have $a_2 = 0$. Then by comparing the coefficients of x^2 and x , we have $a_4 = a_5 = 0$. Finally, by comparing the coefficients of x^0 , we have $a_1 = 0$. Therefore, $a_1 = a_2 = a_3 = a_4 = a_5 = 0$, which shows that the vectors in $\{e_1, e_2, e_3, e_4, e_5\}$ are linearly independent.

(c) Let $W = \text{span}\{e_4, e_5\}$. Since $U = \text{span}\{e_1, e_2, e_3\}$ and notice that $\{e_1, e_2, e_3, e_4, e_5\}$ is a basis of $\mathcal{P}_4(\mathbb{F})$, hence $U \cap W = \{0\}$ and $U + W = \text{span}\{e_1, e_2, e_3, e_4, e_5\} = \mathcal{P}_4(\mathbb{F})$. Therefore, $\mathcal{P}_4(\mathbb{F}) = U \oplus W$.

Problem 8 (2C-8). Suppose $v_1, v_2, \dots, v_m$ is linearly independent and $w \in V$. Prove that

$$\dim \text{span}\{v_1 + w, v_2 + w, \dots, v_m + w\} \geq m - 1.$$

Proof. Denote $v_i + w$ as v'_i and $U = \text{span}\{v_1 + w, \dots, v_m + w\} = U$. We will prove the problem by two steps.

(i) First, we show that the $m - 1$ vectors $v_1 - v_m, v_2 - v_m, \dots, v_{m-1} - v_m$ are linearly independent. Suppose that there exist not-all-zero scalars $a_1, a_2, \dots, a_{m-1}$ such that

$$a_1(v_1 - v_m) + a_2(v_2 - v_m) + \dots + a_{m-1}(v_{m-1} - v_m) = 0.$$

We can rewrite the above equation as

$$a_1v_1 + a_2v_2 + \dots + a_{m-1}v_{m-1} - (a_1 + a_2 + \dots + a_{m-1})v_m = 0.$$

Since $v_1, v_2, \dots, v_m$ are linearly independent, we have $a_1 = a_2 = \dots = a_{m-1} = 0$, which contradicts the fact that the scalars are not all zero. Therefore, the $m - 1$ vectors $v_1 - v_m, v_2 - v_m, \dots, v_{m-1} - v_m$ are linearly independent.

(ii) Next, Note that

$$\begin{aligned} U &\supset \left\{ \sum_{i=1}^m a_i v_i : a_i \in \mathbb{R}, \sum_{i=1}^m a_i = 0 \right\} \\ &= \left\{ \sum_{i=1}^{m-1} a_i v_i - \left(\sum_{i=1}^{m-1} a_i \right) v_m : a_i \in \mathbb{R}, \sum_{i=1}^m a_i = 0 \right\} \\ &= \left\{ \sum_{i=1}^{m-1} a_i (v_i - v_m) : a_i \in \mathbb{R} \right\} \\ &= \text{span}\{v_1 - v_m, \dots, v_{m-1} - v_m\}. \end{aligned}$$

Then By the result in (i), we have

$$\dim U \geq \dim \text{span}\{v_1 - v_m, \dots, v_{m-1} - v_m\} = m - 1.$$

□

Problem 9 (2C-16). Suppose V is finite-dimensional and U is a subspace of V with $U \neq V$. Let $n = \dim V$ and $m = \dim U$. Prove that there exist $n - m$ subspaces of V , each of dimension $n - 1$, whose intersection equals U .

Proof. Take a basis $u_1, \dots, u_m$ of U and extend it to a basis $u_1, \dots, u_m, v_1, \dots, v_{n-m}$ of V . For each $i = 1, \dots, n - m$, let

$$W_i = \text{span}\{u_1, \dots, u_m, v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_{n-m}\}.$$

Then W_i is a subspace of V with dimension $n - 1$ (since it is spanned by $n - 1$ linear independent vectors). We will show that the intersection of these subspaces equals U .

On the one hand, since $U \subset W_i$ for each i , we have $U \subset \bigcap_{i=1}^{n-m} W_i$.

On the other hand, if $v \in \bigcap_{i=1}^{n-m} W_i$, then $v \in W_i$ for each i . Writing v as a linear combination of the basis vectors, we have

$$v = a_1 u_1 + \dots + a_m u_m + b_1 v_1 + \dots + b_{n-m} v_{n-m}.$$

For each $1 \leq i \leq n - m$, $v - b_i v_i$ is a linear combination of $u_1, \dots, u_m, v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_{n-m}$ which are all in V , but $v_i \notin W_i$, consequently we have $b_i = 0$. Therefore, $v = a_1 u_1 + \dots + a_m u_m \in U$. This shows that $\bigcap_{i=1}^{n-m} W_i \subset U$.

Combining the above two parts, we have $\bigcap_{i=1}^{n-m} W_i = U$. $\square$

Problem 10 (3A-4). Suppose $T \in \mathcal{L}(V, W)$ and $v_1, \dots, v_m$ is a list of vectors in V such that $Tv_1, \dots, Tv_m$ is a linearly independent list in W . Prove that $v_1, \dots, v_m$ is linearly independent.

Proof. Suppose that there exist scalars $a_1, \dots, a_m$ such that

$$a_1 v_1 + \dots + a_m v_m = 0.$$

Applying T to both sides of the above equation, we have

$$a_1 Tv_1 + \dots + a_m Tv_m = 0.$$

Since $Tv_1, \dots, Tv_m$ is a linearly independent list in W , we have $a_1 = \dots = a_m = 0$. Therefore, $v_1, \dots, v_m$ is linearly independent. $\square$

Problem 11 (3A-11). Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T is a scalar multiple of the identity if and only if $ST = TS$ for every $S \in \mathcal{L}(V)$.

Proof. On the one hand, if T is a scalar multiple of the identity, then $T = \lambda I$ for some scalar λ . For every $S \in \mathcal{L}(V)$, we have

$$ST = S(\lambda I) = \lambda S = (\lambda I)S = TS.$$

On the other hand, if $ST = TS$ for every $S \in \mathcal{L}(V)$, then we will show that T is a scalar multiple of the identity. Pick a basis $v_1, \dots, v_n$ of V . For each $1 \leq i \leq n$, define the linear map S_i as follow: for arbitrary $v := \sum_{i=1}^n a_i v_i \in V$, $S_i(v) = a_i v_i$. For $i, j \in \{1, 2, \dots, n\}$, set

$$T(v_j) = c_{j1}v_1 + c_{j2}v_2 + \dots + c_{jn}v_n.$$

Then

$$S_i T(v_j) = S_i(c_{j1}v_1 + c_{j2}v_2 + \dots + c_{jn}v_n) = c_{ji}v_i,$$

and

$$TS_i(v_j) = \delta_{ij}T(v_j) = a_j T(v_j) = \delta_{ij}(c_{j1}v_1 + c_{j2}v_2 + \dots + c_{jn}v_n).$$

Since $S_i T = T S_i$, by comparing coefficients, we have $c_{ji} = 0$ for every $i \neq j$. Therefore, for every $1 \leq j \leq n$, there exists a scalar c_j such that

$$T(v_j) = c_j v_j.$$

Next, define S'_i for $2 \leq i \leq n$ as follow: for arbitrary $v := \sum_{i=1}^n a_i v_i \in V$, $S'_i(v) = a_1 v_i$. Then

$$S'_i T(v_1) = S'_i(c_1 v_1) = c_1 v_i, \quad TS'_i(v_1) = T(v_i) = c_i v_i.$$

We immediately have $c_1 = c_i$ for every $2 \leq i \leq n$, which indicates

$$T(v) = c_1 v.$$

□

Problem 12 (3A-13). Suppose V is finite-dimensional. Prove that every linear map on a subspace of V can be extended to a linear map on V . In other words, show that if U is a subspace of V and $S \in \mathcal{L}(U, W)$, then there exists $T \in \mathcal{L}(V, W)$ such that $Tu = Su$ for all $u \in U$.